Any two models of second-order Peano Arithmetic are isomorphic

The Open Logic Project contributors

openlogic:second-order-logic/metatheory:local-snapshot-20261013;
CC BY 4.0

Editorial note. The proof invokes the same-unit standardness theorem for PA2, published as corpus card X2-logic-sol-metatheory-pa2-standard-domain, and the cross-unit standardness of models of Q.

Editorial note. The categoricity claim needs only that isomorphism is an equivalence, which supplies the final composition step.

Editorial scope. One original corollary of the unit's section on the standard model, counted once: categoricity of PA2. The counted corollary and its author proof are retained in full; the paragraph after the proof on PA2-dagger and the proposition with its own completion problem are separate results outside this card.

Allowed standard prerequisites (editorial).

- The second-order Peano axioms PA2: the first eight arithmetical axioms plus the second-order induction axiom, and the notion of a model of a second-order theory.
- The standardness theorem for PA2 of the same unit: the domain of any model of PA2 is exhausted by the values of the numerals; published as corpus card X2-logic-sol-metatheory-pa2-standard-domain.
- The standardness of models of Q proved in the models-of-arithmetic unit: any model of Q whose domain is exhausted by numeral values is isomorphic to the standard natural numbers.
- Every model of PA2 is in particular a model of Q, since Q's axioms are among the arithmetical axioms of PA2.

Source reference. Original theorem sol-pa-standard: sources/olp-second-order-logic-metatheory/pinned-original-source.tex, source lines 638–638. The proof invokes the standardness theorem for PA2 to exhaust the domain of any model by numeral values. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. The theorem is proved in the same unit and is published as corpus card X2-logic-sol-metatheory-pa2-standard-domain.

Source reference. Original proposition thq-standard: sources/olp-model-theory-models-of-arithmetic/pinned-original-source.tex, source lines 1256–1256. The

proof invokes the standardness of models of Q to pass from numeral-exhaustion to isomorphism with the standard naturals. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. The proposition is proved in the models-of-arithmetic unit of the same source tree.

Corollary 0.1. *Any two models of* ThPA^2 *are isomorphic.*

Proof. By sol-pa-standard , the domain of any model of ThPA^2 is exhausted by val$(\overline{n})M$. Any such model is also a model of ThQ. By thq-standard , any such model is standard, i.e., isomorphic to $\mathfrak{N}$. □